Client-side energy budget of the GTSRB CNN

comb vs. time-serial encoding, in the Eq. (5) convention of Sec. II D

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Convention (paper, Sec. II D / Methods G)

$e_{\mathrm{ip}} = e_1 + e_2 + e_3$ with $e_1 = P_x T_{\mathrm{air}}/(4N\,\eta_{\mathrm{radio}})$ (transmit; P_x *at the mixer*, $\eta_{\mathrm{radio}} = 10\%$), $e_2 = 6\,e_{\mathrm{ADC}}/4N$, $e_3 = 8\,e_{\mathrm{dig}}/4N$ ($e_{\mathrm{ADC}} = e_{\mathrm{dig}} = 1\,\mathrm{pJ}$; fixed 14 pJ read-out fee per answer). Component-level and input-referred on both sides: no DAC, no static transceiver power, no SDR wall-plug; H100 reference 70 fJ per real MAC (arithmetic only). One complex inner product of length $N = 4N$ real MACs.

1. Reproduction of Sec. II D (resolved inner products)

	P_{LO}	P_{RF} at mixer	e_1	fee (e_2+e_3)	e_{ip}	paper
$N = 4096$	-3 dBm	$0.526\,\mathrm{nW} = -62.8\,\mathrm{dBm}^{\dagger}$	0.616 fJ	0.854 fJ	**1.47 fJ**	1.47 fJ
$N = 65{,}536$	-3 dBm	$0.460\,\mathrm{nW} = -63.4\,\mathrm{dBm}$	0.538 fJ	0.053 fJ	**0.59 fJ**	0.59 fJ

Input powers: weight comb fixed at $P_{LO} = -3\,\mathrm{dBm}$ (paper, Methods C); data-comb P_{RF} closed-loop tuned to the 25-dB guard-bin target and frozen (paper) — tabulated above. GPU crossover $N \approx 50$ (paper: $N \gtrsim 50$); radio-floor bend $N \approx 6503$ (paper: 6.5×10^3); 48× / 119× below H100 (paper: 48×/119×). *All Sec. D numbers reproduce.* †backed out from the printed 1.47 fJ (P_x for N=4096 is not stated in the draft). Open item: 117 ns airtime per real MAC is ∼13% above the bare frame arithmetic $(L+\mathrm{CP})/4N = 103\,\mathrm{ns}$ at 10 MS/s — presumably sync amortization; to be confirmed against the original generator.

2. Comb CNN at $(P_{LO}, P_{RF}) = (-3, -65)\,\mathrm{dBm}$ at the mixer

The data comb is inserted at the mixer RF port at -65 dBm — the operating point of the frozen-battery demonstration as physically delivered. Client pays the data comb only (weights = base-station side). Symbol counts from the fielded planner (`plan_layer`, as in `miwen_overhead_ratio.py`); r3plus: 7.21 M complex MACs = 28.85 M real MACs, 31,659 answers, 1,628 symbols (+3.1% sync) → **2.84 s airtime/image**.

layer	D	sym/img	cMAC/img	answers	fee per real MAC
conv1	75	784	1,881,600	25,088	46.7 fJ
conv2	800	800	5,120,000	6,400	4.4 fJ
dense1	1600	43	204,800	128	2.2 fJ
dense2	128	1	5,504	43	27.3 fJ

	per image	per real MAC
e_1 (TX at $-65\,\mathrm{dBm} = 0.316\,\mathrm{nW}$)	9.0 nJ	0.31 fJ
read-out fee (14 pJ × 31,659)	443.2 nJ	15.36 fJ
total	**452.2 nJ**	**15.68 fJ** — 4.5× below H100

The NN budget is *read-out–fee dominated*, the opposite of Sec. D's large-N points: the CNN's inner products are short ($D = 75 \ldots 1600$), so the fixed 14 pJ/answer amortizes poorly (conv1 pays 46.7 fJ per real MAC by itself). The comb packing is not the problem — airtime works out to 98 ns per real MAC, essentially Sec. D's 117 ns.

3. Serial CNN at $(P_{LO}, P_{RF}) = (-10, -24)$ dBm

The measured low-drive serial arm (`serial_nn_runner_m10m24.py`, unpadded chain → commanded ≈ device plane). One complex product per 3.2 μs slot: 7,211,904 slots → **23.1 s airtime/image**. Read-out fee assumes the slow coherent integrate-and-dump read-out at the *row* rate (Sec. D convention); the as-fielded per-sample SDR capture would instead cost 16.0 pJ per real MAC.

	per image	per real MAC
e_1 (TX at -24 dBm $= 4.0\,\mu$W)	918.8 μJ	31.85 pJ
read-out fee	443.2 nJ	15.36 fJ
total	**919.2 μJ**	**31.86 pJ** — 455× *above* H100

Bottom line

Sec. D reproduces exactly (1.47 / 0.59 fJ per real MAC, crossover $N \approx 50$, bend $N \approx 6.5{\times}10^3$). **Comb CNN at** $(-3, -65)$ dBm**: 452 nJ/image = 15.7 fJ per real MAC, 4.5× below the H100 line** — but unlike the large-N inner products the budget is 98% read-out fee, because the network's inner products are short; the radio itself is nearly free (0.31 fJ). **Serial CNN at** $(-10, -24)$ dBm**: 919 μJ/image = 31.9 pJ per real MAC, 455× above H100** — serial's *radio* term is $\sim 10^5\times$ the comb's (+41 dB power, -24 vs. -65 dBm, × 8× airtime per MAC, 0.8 μs vs. 98 ns); the *totals* differ 2,000× because both pay the same read-out fee, which dominates the comb budget. At the headline $(0, 0)$ dBm serial point e_1 grows another +24 dB (≈8 nJ per real MAC). Frequency multiplexing, not the mixer, is what makes the NN energy story work.

Assumptions (knobs in the code): sync overhead 1/32; one average per symbol; capture/re-tune gaps excluded; client pays the data-comb port only; Sec. D exclusions throughout. Numbers tagged [paper]/[repo]/[derived] in `energy_budget_nn.py`.